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Name : Kranti Sunil Dhanawade Subject : IIOT.
Std : BE Div. : Roll No : A-30
~~School~~ / College : GSMCOE

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Practical Assignment No - 1

Title :- Sending alert messages to the user for controlling and interacting with environment.

Aim :- write a program that sends messages to the user for controlling and interacting with environment.

main objective : Design program that simulates sending alert messages to the user based on environmental conditions.

Prerequisites :- C/C++ Programming

Requirements :- Software : Arduino IDE

Learning objective : Learn to Environmental monitoring, conditional logic, event handling similated sensor data, alert notifications.

Outcome :- understand how to create effective alert system, understanding of sensor data simillation.

Theory :-

Gas Sensor

Gas sensors are tuned to detect the presence of combustile toxic, or inflammable gas in the vicinity of the sensor. Gas sensors are devices that detect the presence of gases and measure their concentration by physical or chemical interactions into electrical signals.

Measuring Gases

when measuring gases such as CO_2, O_2, methane, but the quantity of gas per volume of air is called concentration.

The ratio of one gas to another is given in parts per million.

pppm = 1/10,00,000 = 0.0007%

Procedure :-

1. connect the components to arduino
2. open Arduino IDE and write the program.

3. Select the correct board and com. port.

4 Upload the program and open serial monitor at 9600 baud.

5 Observe the sensor reading ~~read~~ LED and buzzer.

Observation :- Below the threshold, the LED and buzzer remain off. Above the threshold, the LED turns ON and the buzzer rings.

Conclusion : An alert system using an MQ-2 gas sensors was implemented successfully.

Experiment : 2

Interfacing with PIR sensor.

Aim :- To Interface a PIR sensor with Arduino for motion detection.

Theory :-

A passive Infrared (PIR) sensor detects changes in infrared radiation caused by moving object and provides a digital output.

Apparatus :- Arduino uno, PIR sensor, jumper wires and USB cable.

Connections :

PIR VCC - 5V
PIR GND - GND
PIR OUT - D13

Procedure :-

1. connect the PIR sensor to Arduino
2. Open Arduino IDE and write the program.

3. Select the correct board and com port.
4. Upload the program.
5. Open Serial Monitor at 115200 baud
6. Move in front of the sensor and observe the output.

Observations:

No movement : No motion Detected

Movement : Motion Detected.

Conclusion:

The PIR sensor was Interfaced with Arduino successfully.

Experiment 3:

monitoring and controlling a CPS System.

Aim :- To design a user interface for monitoring and controlling a cyber-physical-system (CPS).

Theory :-

A CPS integrates physical device, sensors and computer-based monitoring and control. It collects real-time data and allows users to control devices through a user interface.

Apparatus :- Arduino Uno, MQ-2 gas sensor, DS18B20 temp sensor, HC-SR04 ultrasonic sensor, LED, buzzer 220Ω resistor, jumper wires and USB cable.

Connections.

MQ-2 :- VCC - 5V, GND - GND, DOUT - D13 AOUT - A5.

LED :- Anode - D12 through resistor, Cathode - GND.

Buzzer :- Positive - A0, Negative - GND

Procedure:

1. Connect all sensors and devices to Arduino.
2. Write and upload the program using Arduino IDE
3. Open Serial Monitor at 9600 baud.
4. Observe gas, temp & distance readings
5. Monitor sensor status through the CPS interface.
6. Control the LED and buzzer and start/stop real-time monitoring.

Observations:

The system displays gas, temperature and distance readings. The interface allows monitoring and control of the LED and buzzer.

Conclusion:

A CPS monitoring and controlling system was implemented successfully.

Experiment 4.

Implementing security measures in an IIOT system.

Theory:

Aim :- To Implement security measures in an IIOT system.

Theory :-

IIOT security protects connected industrial devices and networks from unauthorized access and cyber threats. Authentication and limited logic attempts help prevent unauthorized access.

Apparatus :- Arduino Uno, PIR sensor, HC-SR04 ultrasonic sensor, LED, buzzer, 220Ω resistor, jumper wires and USB cable.

Connections:-

PIR :- VCC - 5V, GND - GND, OUT - D13.

HC-SR04 :- VCC - 5V, GND - GND

LED: Anode :- D-12 through resistor, cathode - GND.

Buzzer :- Positive - A0, Negative - GND

Procedure :

1. connect the sensors, LED and buzzer to Arduino.
2. Write and upload the security program.
3. Open serial monitor at 9600 baud.
4. Trigger motion using the PIR sensor and measure distance using HC-SR04.
5. Enter the username and password.
6. Observe access granted or system lockout after failed attempts.

Observation :

Correct credentials allow accey. Invalid usernames or repeated Incorrect passwords trigger a security alert through the LED and buzzer.

Conclusion :- Security measures for an IIOT system was Implemented successfully.

Experiment 5

Industrial Data Analysis

Aim : To perform Industrial data analysis using relevant tools and techniques.

Theory

Industrial data analysis involves collecting, cleaning, processing and analyzing sensor data to improve Industrial performance. Excel, Python and Visualization tools help identify patterns and trends.

Apparatus:-

Arduino Uno, LDR sensor, potentiometer, DHT22 temperature and humidity sensor, jumper wires and USB cabel.

Connections :.

LDR module :- VCC - 5V, GND - GND, DO - D13

DHT22 :- VCC - 5V, GND - GND, SDA - A4.

Potentiometer : VCC - 5V, GND - GND, SIG - A0

Procedure.

1. connect the sensor to Arduino
2. write and upload the program.
3. open serial monitor at 9600 baud.
4. vary light intensity using the potentiometer and observe readings.
5. Record temp & humidity readings from different environment.
6. Import the data into Excel and clean it.
7. Calculate average, maximum and minimum values.
8. Create line and bar charts for visualization.

Observation :-

LDR readings change with light intensity. The DHT22 records temp and humidity. The

collected data is cleaned, analyzed and visualized using Excel.

Conclusion:-

Industrial sensor data was collected, analyzed and visualized succesfully.